

Ruitian Liu

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[GitHub](#) ↗

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Education

University of California, Davis

2025.9 - 2027.6 (Expected)

M.S. in Statistics GPA: 3.70

Relevant Coursework: Computational Statistics (Ph.D level), Machine Learning (Ph.D level), Object Data Analysis (Ph.D level, A), Intensive Longitudinal Data Analysis (Ph.D level, A), Bayesian Data Analysis (A+)

North China Institute of Science and Technology

2021.9 - 2025.6

B.S. in Statistics GPA: 3.59 Major GPA: 3.82

Relevant Coursework: Mathematical Analysis I/II/III, Linear Algebra I/II, Analytical Geometry, ODE, Multivariate Statistics (A+), Sampling Theory (A+), Stochastic Processes, Time Series Analysis, Nonparametric Statistics

- First-Class Outstanding Graduate

Academic Experience

Functional Representation and Regression of Typhoon Evolution

[2026.1 – 2026.3]

Supervised by Hans-Georg Müller

- Applied FPCA to 1,101 typhoon intensity trajectories from NOAA IBTrACS; three principal components explain over 90% of variation, capturing overall intensity, peak timing, and short-term oscillation.
- Regressed FPC scores on storm-level predictors and fit concurrent varying-coefficient models on spatial displacement, using functional concurrent model identifying time-varying latitude/longitude effect on storm intensity.

Causal Re-Analysis of Project STAR: Do Small Classes Really Help?

[2026.1 – 2026.3]

Supervised by Prof. Shizhe Chen

- Re-examined Tennessee's Project STAR under differential attrition and post-baseline class switching, and redefined the estimand using kindergarten ITT assignment.
- Built a pipeline combining MICE, doubly robust AIPW, and multilevel mixed models; estimated a +7.5 point kindergarten math effect and small-class effect stays roughly similar across grades

Benchmarking Interpolation Methods for Irregularly-Sampled AR(1) Processes

[2025.11 – Present]

Supervised by Prof. Siwei Liu

- Benchmarked six interpolation methods on irregular AR(1) series under varying noise, missingness, dependence, and nonstationarity, jointly evaluating reconstruction error, shape recovery, and bias in downstream $\hat{\phi}$ estimation.
- No method dominated: GRU performed well in difficult regimes but was more costly and less robust to nonstationarity; probabilistic methods handled noise better, while splines balances between robustness and flexibility.

Publications

Zhang, D., Yang, W., Jiang, H., Ruitian Liu, Kong, D. "Research on urban emergency material support capacity based on segmented driven three-way decision VIKOR method." *Journal of the Operational Research Society*, 2026.

- Contributed to the robustness analysis: designed perturbation experiments on indicator extremes, means, and dispersion, and ran randomized trials comparing different methods.

Teaching Experience

Statistics Tutor, UC Davis AATC

2026.2 – Present

- STA 013 Elementary Statistics
- STA 100 Applied Statistics for the Biological Sciences
- SOC 056 Introductory Social Statistics

Awards

- National Scholarship, Ministry of Education, China, 2024

Skills

- Programming: R (including Rcpp), Python, Stan, Matlab, LaTeX